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INSTITUTE OF SCIENCE, TECHNOLOGY & ADVANCED STUDIES (VISTAS)
(Deemed to be University Estd. u/s 3 of the UGC Act, 1956)
PALLAVARAM, THALAMBUR, PERIYAPALAYAM, THIRUVANMIYUR, ADYAR - CHENNAI



AI-ASSISTED EV RANGE PREDICTION AND SMART CHARGING OPTIMIZATION SYSTEM

FINAL YEAR

PROJECT PHASE-I REPORT

Submitted by

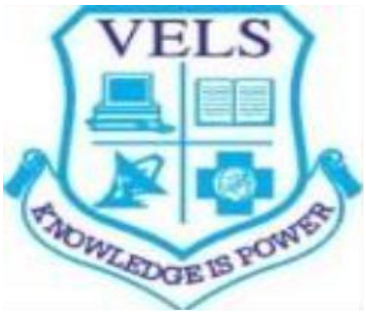
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in Partial fulfillment for the award of the degree of

BACHELOR OF TECHNOLOGY CSE - ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING



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VELS INSTITUTE OF SCIENCE, TECHNOLOGY AND ADVANCED STUDIES (VISTAS)
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BONAFIDE CERTIFICATE

Certified that this project report **"AI-ASSISTED EV RANGE PREDICTION AND SMART CHARGING OPTIMIZATION SYSTEM"** is the bonafide work of **Shoulina A (23619231), Sanjai S (23619229), Madhumitha M (23619242)** who carried out the project under our supervision. Certified further that, to the best of our knowledge, the work reported here does not form part of any other project report or dissertation on the basis of which a degree or award was conferred earlier on this or any other candidate

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AI-ASSISTED EV RANGE PREDICTION AND SMART CHARGING OPTIMIZATION SYSTEM

ABSTRACT

This project develops a Chennai-focused EV trip-planning system that combines route planning, XGBoost energy prediction, battery SOC/range estimation and reserve-constrained charging optimization. Users select locations, vehicle details and a route. The system predicts trip energy, estimates arrival SOC and evaluates charging needs under a 25% planning reserve.

OBJECTIVES

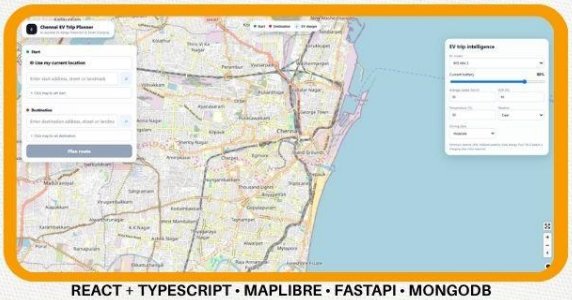
- Generate primary and alternative routes.
- Predict trip energy using XGBoost.
- Estimate arrival SOC and safe range.
- Filter reachable charging stations.
- Optimize charging quantity and planning cost/time.

METHODOLOGY

- Location input and Chennai GIS/PIN resolution
- OSMnx + NetworkX road graph; Dijkstra routes
- User selects a preferred route
- XGBoost predicts trip energy
- SOC and safe-range calculation
- 25% reserve check; station filtering; PuLP/MILP

Workflow: Location → Route → Energy → SOC/Reserve → Charging → Trip Plan

IMPLEMENTED PLATFORM

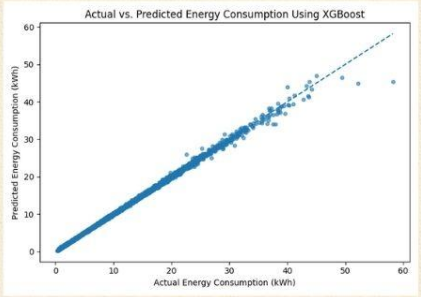


REACT + TYPESCRIPT • MAPLIBRE • FASTAPI • MONGODB

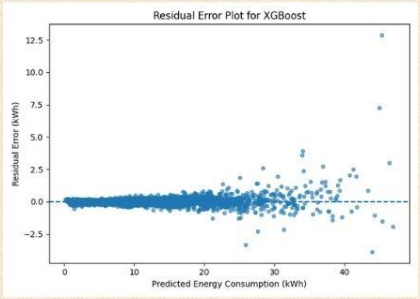
RESULTS & DISCUSSION

R^2	MAE	RMSE
0.9978	0.1481	0.3613

The actual-versus-predicted graph shows most points close to the ideal line. Residuals are mainly near zero, with some larger errors at higher energy values.



ACTUAL VS. PREDICTED ENERGY



RESIDUAL ERROR PLOT

CONCLUSION

The integrated application links route selection, energy prediction, SOC estimation and charging optimization. The test metrics describe performance on the available dataset; independent real-world fleet validation remains future work.

STANDARDS & TECHNICAL CONVENTIONS

- JSON API payloads: RFC 8259.
- GeoJSON geographic data convention: RFC 7946; coordinates are longitude, latitude.
- PEP 8 used as a Python code-style reference; no formal certification is claimed.

ABSTRACT

Electric vehicle trip planning faces challenges to uncertainty in energy use, battery levels, road conditions and charging needs. This paper introduces an AI-driven system for predicting EV range and optimizing charging. The system combines route planning, machine learning for energy prediction, battery state tracking and charging station selection using mathematical optimization. It works within the Chennai District using browser-based tools for location input, like geolocation address search and map selection. A road network built from OpenStreetMap data enables route calculations. Dijkstra's algorithm finds the path with alternate routes available for users to choose. An XGBoost regression model predicts energy use based on vehicle details, battery condition, road type, speed, weather and driving behavior. The model's output is used with the starting battery state of charge to calculate energy, driving range and arrival SOC ensuring a minimum 25% battery reserve. If the trip exceeds the range the system identifies nearby charging stations that are reachable. It then uses a PuLP-based mixed-integer linear programming model to decide when and where to charge. The system is built with React, MapLibre, FastAPI, MongoDB, Python, OSMnx, NetworkX, XGBoost and PuLP. Testing with 20,000 real EV trip records showed performance: XGBoost achieved an R^2 of 0.9978, a MAE of 0.1481 and RMSE of 0.3613.

Keywords—Electric Vehicles, Energy Consumption Prediction, XGBoost State of Charge Driving Range, Dijkstra Algorithm, Charging Station Selection, Smart Charging, Mixed-Integer Linear Programming, PuLP, Route Planning, Geospatial Computing.

CHAPTER 1
INTRODUCTION

1.1 GENERAL

Electric vehicles are increasingly used as an alternative to conventional vehicles powered by petrol or diesel. They use rechargeable batteries to supply energy to an electric motor. The practical distance an EV can travel depends on battery capacity, the remaining charge, vehicle characteristics, road conditions and driving behaviour.

Modern EVs generally provide a battery state-of-charge indicator and an estimated remaining range. However, the energy required for a particular journey may change according to the route selected, distance travelled, vehicle model, temperature, road type and driving style. A trip-planning application can therefore provide additional decision support by estimating energy consumption for the selected route and checking whether the destination can be reached while maintaining a minimum battery reserve.

The proposed project, **AI-Assisted EV Range Prediction and Smart Charging Optimization System**, integrates route planning, energy prediction, SOC estimation and charging optimization. It is designed for Chennai District and provides a browser-based interface through which users can enter locations, choose an EV model and examine route alternatives.

The system uses geographic information to process locations and construct routes. The selected route is passed to an XGBoost model that predicts the required trip energy. This prediction is combined with the initial SOC and battery capacity to estimate arrival SOC and safe range. If the estimated arrival SOC falls below the configured 25% reserve, the system checks charging stations and attempts to identify a feasible charging plan.

The project is a software-based trip-planning and decision-support prototype. It does not replace the vehicle's Battery Management System (BMS), directly measure the physical battery state or automatically control an EV charger. Its purpose is to combine prediction, routing and charging decisions in one application.

1.2 PROBLEM STATEMENT

Although modern electric vehicles provide SOC and estimated driving range, trip-level energy requirements and charging decisions can vary with the selected route, vehicle, battery and driving conditions. A driver planning a particular journey needs to understand whether the selected route can be completed while maintaining a practical battery reserve and, if not, which charging option is feasible.

Selecting the nearest charging station alone may not produce a suitable recommendation. The station must be reachable using the available battery energy, and the charging decision must consider the required charge amount, estimated charging duration and cost. These factors need to be evaluated together with the selected route and predicted energy consumption.

Therefore, the problem addressed by this project is the development of an integrated EV trip-planning application that predicts trip energy, estimates arrival SOC and safe range, checks a minimum SOC reserve and recommends charging when a feasible option exists.

The system combines machine learning with route processing and mathematical optimization. It uses XGBoost for energy prediction and a PuLP-based MILP model for charging decisions. If no candidate station satisfies the configured constraints, the system reports that no feasible station was found instead of forcing an unsuitable recommendation.

1.3 OBJECTIVE

The main objective of this project is to develop an AI-assisted EV trip-planning application that estimates trip energy consumption and supports charging decisions based on the selected route and battery reserve.

The specific objectives are:

1. To develop a web interface for entering origin, destination, EV model and initial battery SOC.
2. To process geographic locations and generate a primary route with alternative routes.
3. To predict trip energy consumption using an XGBoost regression model.
4. To estimate arrival SOC and safe driving range from predicted energy and battery information.
5. To apply a minimum 25% SOC reserve as a trip-planning constraint.
6. To identify charging stations that satisfy the modeled route and reachability requirements.
7. To use PuLP and Mixed-Integer Linear Programming to determine a feasible charging station and charging amount when possible.
8. To display route information, predicted energy, battery estimates and charging recommendations through a web dashboard.
9. To evaluate the prediction model using R^2, MAE and RMSE and test the integrated workflow using representative trip scenarios.

1.4 MODULES

The system is divided into five main modules. Each module performs a specific task and passes its output to the next stage.

Module 1 – Energy Prediction

This module uses the selected EV details, battery information, route distance and operating-condition features to predict trip energy consumption. The XGBoost regression model returns an estimated energy value in kilowatt-hours (kWh).

Module 2 – Range and SOC Estimation

This module combines the predicted trip energy with the initial SOC and battery capacity. It calculates estimated remaining energy, arrival SOC and safe range. The module checks whether the estimated arrival SOC satisfies the configured 25% minimum reserve.

Module 3 – Charging Station Selection

When the predicted trip does not maintain the minimum reserve, this module identifies charging stations related to the selected route. It checks whether candidate stations satisfy the implemented reachability and reserve conditions before passing feasible candidates to the optimizer.

Module 4 – Smart Charging Optimization

This module uses PuLP to formulate a Mixed-Integer Linear Programming model. It evaluates eligible station choices and charging quantities while considering the modeled charging cost, charging time and battery constraints. If no feasible solution exists, the system reports that outcome.

Module 5 – Trip Monitoring and Dashboard

This module displays the selected route, route distance, travel duration, predicted energy, estimated SOC, safe range and charging status. When a feasible charging plan is available, the dashboard also displays the recommended station and estimated charging details.

The five modules operate as a connected workflow. Energy prediction provides the input for SOC estimation, the SOC reserve check controls whether charging analysis is required, and the final dashboard presents the result.

1.5 SUMMARY

This chapter introduced the need for trip-level EV energy estimation and charging decision support. The project addresses the challenge of connecting a selected route with predicted energy consumption, arrival SOC, safe range and charging feasibility.

The proposed system combines geographic routing, XGBoost energy prediction and PuLP-based charging optimization. It applies a 25% SOC reserve as a planning rule and displays the resulting trip plan through a web interface. The next chapter reviews existing research related to EV energy prediction, route planning and charging-station optimization.

CHAPTER 2

LITERATURE SURVEY

2.1 GENERAL

Electric-vehicle (EV) trip planning has become an important research area due to the increasing adoption of electric mobility. Unlike conventional vehicles, electric vehicles depend on stored battery energy, which must be managed carefully throughout a journey. Factors such as distance, road conditions, vehicle speed, traffic, elevation and driving behaviour influence energy consumption. Therefore, effective EV trip planning requires the integration of route optimization, energy prediction, battery-state estimation and charging-station recommendation.

Energy-consumption prediction can be performed using physical models, statistical methods, machine-learning algorithms and deep-learning techniques. The suitability of each method depends on the available dataset, input features, prediction accuracy and computational requirements. Machine-learning approaches can identify relationships between vehicle characteristics and energy usage from historical data.

Graph algorithms are widely used to identify routes between locations. Dijkstra's algorithm determines the shortest path when the graph has non-negative edge weights. Alternative-route methods allow users to compare different paths according to distance or other configured criteria. For electric vehicles, route information can be combined with energy prediction to estimate the battery state at the destination.

Charging-station optimization considers factors beyond geographical distance. A suitable recommendation should account for station reachability, remaining battery energy, minimum SOC reserve, charging quantity, charging duration and estimated cost. Mathematical optimization techniques can represent station selection and charging quantities as decision variables subject to operating constraints.

The literature provides a foundation for combining machine learning, graph-based routing and mathematical optimization. The proposed system integrates these functions into a single web application that supports EV trip planning within Chennai District.

2.2 EXISTING SYSTEM

Existing EV trip-planning systems generally provide functions such as battery state-of-charge monitoring, remaining-range estimation, navigation, charging-station search and charging recommendations. Vehicle manufacturers provide battery information through vehicle displays or companion applications, while navigation services offer route directions and charging-location information.

Conventional navigation systems primarily focus on distance or estimated travel time. However, the shortest route may not always be the most energy-efficient route because road gradients, traffic

conditions and driving speeds affect energy consumption. Consequently, route distance alone may not accurately represent the energy required for an electric vehicle.

Battery-monitoring systems display the current SOC, while range-estimation systems calculate the approximate distance that can be travelled using the remaining energy. However, these estimates may not fully account for the specific road conditions of a selected route. An inaccurate energy estimate can lead to unnecessary charging stops or an unsuitable trip plan.

Charging-station applications help users locate nearby charging infrastructure. Nevertheless, a nearby station may not be reachable while maintaining the required battery reserve. Station information may also lack real-time details about connector availability, operating status and charging prices.

Research approaches have examined individual aspects of EV planning, including energy prediction, route generation and charging optimization. However, their methods, datasets and assumptions may differ, making direct comparisons difficult.

The proposed system addresses the integration of these functions by combining route planning, data-driven energy prediction, SOC estimation and reserve-constrained charging optimization. It is intended to complement existing navigation and battery-monitoring systems rather than replace them.

2.3 PROPOSED SYSTEM

The proposed system is an AI-assisted EV trip planner designed for Chennai District. The user enters the origin and destination, selects an EV model and provides the initial SOC and other available operating parameters. The application resolves the locations and generates a primary route along with alternative routes for comparison.

The route-processing module uses OpenStreetMap-derived road data with OSMnx and NetworkX. The road network is represented as a graph containing nodes and connecting road segments. Dijkstra's algorithm generates the primary path according to the configured edge weights, while the application provides alternative routes for user selection.

After route selection, the backend prepares the required input features and uses XGBoost to predict trip energy consumption. The predicted energy is passed to the battery-state module, which estimates the arrival SOC and remaining range.

A minimum 25% SOC reserve is used to determine whether the trip is feasible without intermediate charging under the model assumptions. If the estimated arrival SOC falls below this reserve, the application identifies candidate charging stations and evaluates their reachability along the selected route.

The charging module uses a mixed-integer linear programming (MILP) model implemented with PuLP. It evaluates station-selection decisions and charging quantities according to the configured charging cost or time objective and the constraints represented in the model. If no feasible station combination satisfies these constraints, the system reports the infeasible result.

The frontend is developed using React and TypeScript, while MapLibre GL JS displays the map, routes and charging locations. FastAPI and Python coordinate backend processing, and MongoDB supports data storage. The dashboard presents the selected route, predicted energy consumption, estimated arrival SOC and charging recommendations.

The main contribution of the proposed system is the integration of route planning, energy prediction, SOC estimation and charging optimization within one browser-based application.

2.3.1 Route Planning

Route planning determines a suitable path between the origin and destination using road-network data. OpenStreetMap provides geographic road information, OSMnx supports road-network retrieval and processing, and NetworkX provides graph algorithms.

Dijkstra's algorithm identifies the minimum-cost path according to the assigned edge weights. When the weights represent distance, it returns a shortest-distance path. The resulting route is then used by the application to support energy prediction and trip feasibility analysis.

Route quality depends on the accuracy of the mapped road network, road connectivity and the chosen routing objective. Therefore, the generated route represents a planning estimate rather than a guarantee of actual driving conditions.

2.3.2 Energy Prediction Using XGBoost

XGBoost is a gradient-boosting framework that combines decision trees to produce predictions. It can model nonlinear relationships between input features and a continuous target, making it suitable for energy-consumption prediction when appropriate training data is available.

In the proposed system, XGBoost estimates the energy required for the selected trip using the features supported by the trained model. The prediction is subsequently used to calculate the expected arrival SOC and determine whether charging is required.

The reliability of the prediction depends on the training dataset, feature selection, data preprocessing and model evaluation. Metrics such as Mean Absolute Error (MAE), Root Mean Squared Error (RMSE) and R-squared can be used to evaluate prediction performance using an independent test dataset.

2.3.3 SOC and Range Estimation

State of charge represents the remaining battery charge as a percentage. The estimated arrival SOC is calculated using the SOC, battery capacity and predicted energy consumption.

A simplified relationship is:

$$SOC_{arrival} = \frac{E_{remaining}}{C_{battery}} \times 100$$

where (E) represents predicted energy consumption in kWh and (C) represents battery capacity in kWh.

The application uses a minimum 25% SOC reserve to evaluate trip feasibility. If the predicted arrival SOC is below this threshold, the system evaluates the need for intermediate charging.

Remaining range is estimated from the available usable energy and expected energy consumption per kilometre. These calculations provide useful planning estimates, but their accuracy depends on the energy model and operating assumptions.

2.3.4 Charging-Station Optimization

Charging-station optimization identifies suitable charging opportunities when the available battery energy is insufficient to complete the trip while maintaining the required reserve.

Candidate stations are evaluated using the route and reachability constraints implemented in the application. Where supported, additional considerations may include charging power, connector compatibility and station availability.

PuLP is used to formulate the charging problem as a MILP model. Binary decision variables can represent whether a station is selected, while continuous variables represent charging quantities. The objective minimizes the configured charging cost or time subject to the modeled energy and feasibility constraints.

The optimizer returns a charging plan only when the formulated problem has a feasible solution. If no solution satisfies the constraints, the system reports infeasibility rather than presenting an invalid recommendation.

2.3.5 Frontend and Backend Integration

React and TypeScript are used to develop the user interface, allowing users to enter trip details, select routes and view prediction results. MapLibre GL JS displays geographic information, route paths and charging-station markers.

The backend uses FastAPI and Python to coordinate route processing, model inference, SOC calculations and charging optimization. MongoDB supports the storage of application data according to the implemented database structure.

The dashboard presents the main trip-planning results, including route information, predicted energy consumption, arrival SOC and charging recommendations. Clear presentation helps users distinguish between entered values, calculated estimates and optimization outputs.

2.4 COMPARATIVE ANALYSIS

Existing navigation systems focus mainly on route generation, while battery-monitoring systems provide SOC information and charging applications identify charging locations. These functions are useful individually but may not provide a unified assessment of route feasibility and charging requirements.

The proposed system combines these functions into one workflow. Dijkstra's algorithm generates the primary route, XGBoost predicts energy consumption, the SOC module estimates the remaining battery level, and the PuLP-based MILP model evaluates charging decisions.

The integration allows the selected route to be assessed using predicted energy demand and a predefined battery reserve. However, the system's practical performance must be established through testing and comparison with appropriate baseline methods

2.5 RESEARCH GAPS IDENTIFIED

The review identifies several challenges in EV trip planning.

1. Separation of route planning and energy prediction: A shortest-distance route may not minimize energy consumption. Connecting route selection with energy prediction helps evaluate the expected battery requirement of the selected path.

2. Battery reserve management: A trip may be reachable while leaving an insufficient battery reserve. The proposed system uses a minimum 25% SOC reserve as an explicit planning constraint.

3. Charging quantity estimation: Identifying a charging station does not determine the energy required for the remaining journey. The optimization module evaluates charging quantities according to the modeled trip requirements.

4. Feasibility checking: Some trips may not be feasible with the available battery capacity and charging infrastructure. Explicit infeasibility handling prevents the system from presenting an unsupported charging plan.

5. Dataset limitations: Machine-learning accuracy depends on representative training data, suitable input features and independent testing. The energy-prediction model must be evaluated before its results can be treated as reliable.

6. Geographic data reliability: Road-network inaccuracies and incomplete charging-station information can affect recommendations. Appropriate geographic data and validation are necessary for dependable operation.

2.6 SUMMARY

This chapter reviewed the existing approaches used in EV energy prediction, route planning, SOC estimation, range calculation and charging-station optimization. Although these approaches provide important individual capabilities, an effective trip-planning application must connect them to produce consistent decisions.

The proposed system integrates OSMnx and NetworkX for road-network processing, Dijkstra's algorithm for primary-route generation, XGBoost for energy prediction, SOC calculations for battery-state estimation and PuLP/MILP optimization for charging decisions. React, TypeScript and MapLibre GL JS support the frontend, while FastAPI, Python and MongoDB support backend processing and data storage.

The system applies a 25% SOC planning reserve and evaluates charging requirements when the predicted arrival SOC is insufficient. Its main contribution is the integration of route planning, energy prediction and reserve-constrained charging optimization within a single browser-based application.

The accuracy and usefulness of the results depend on the quality of the training dataset, geographic information, station data and optimization constraints. The next chapter explains the system requirements, architecture, modules, dataset preparation and proposed methodology.

CHAPTER 3

3.1 METHODOLOGY:

3.1.1 SYSTEM REQUIREMENT

The system requires hardware and software capable of running a browser-based interface, backend API, road-network processing, machine-learning inference and mathematical optimization.

Hardware Requirements

- A laptop or desktop computer.
- A modern web browser.
- Sufficient RAM and storage for the application, dataset, geographic data and model files.
- An internet connection when accessing online map resources or external geographic services.

The prototype does not require dedicated EV sensors for its normal software-based trip-planning workflow. It uses user-provided parameters and structured dataset features rather than directly measuring a physical vehicle's battery state.

Software Requirements

Component	Technology	Purpose
Frontend	React and TypeScript	User interface and trip input
Mapping	MapLibre GL JS	Map and route visualization
Backend	FastAPI and Python	API and application processing
Routing	OSMnx and NetworkX	Road graph and path computation
Energy prediction	XGBoost	Trip-energy regression
Optimization	PuLP / MILP	Charging decision optimization
Database	MongoDB	Document and geospatial data storage
Geographic data	OpenStreetMap, GCC GIS and PIN data	Route and location processing

3.2 SYSTEM ARCHITECTURE OF THE PROPOSED MODEL

The system architecture consists of five connected layers: user interface, backend API, location and route processing, energy and battery analysis, and charging optimization.

The user interface is implemented using React and TypeScript, with MapLibre GL JS displaying the map. The user provides the trip locations and vehicle parameters. These inputs are sent to the FastAPI backend for processing.

The location and route layer resolves geographic coordinates and connects them to the road graph. OSMnx and NetworkX process the road network, while Dijkstra's algorithm generates the primary route. Alternative routes are supplied for user selection.

The energy-prediction layer prepares the selected route and EV information as model inputs. XGBoost predicts the required trip energy. The battery-estimation layer combines the prediction with initial SOC and battery capacity to calculate estimated arrival SOC and safe range.

The charging layer checks the 25% reserve. If the trip is feasible without charging, the result is sent to the dashboard. Otherwise, charging stations are filtered for route relevance and reachability. PuLP/MILP evaluates feasible charging decisions, and the final result is returned to the interface. MongoDB supports storage and retrieval of project and geographic records. The architecture keeps frontend presentation, backend orchestration, route processing, model inference and optimization as separate responsibilities.

System architecture flow:

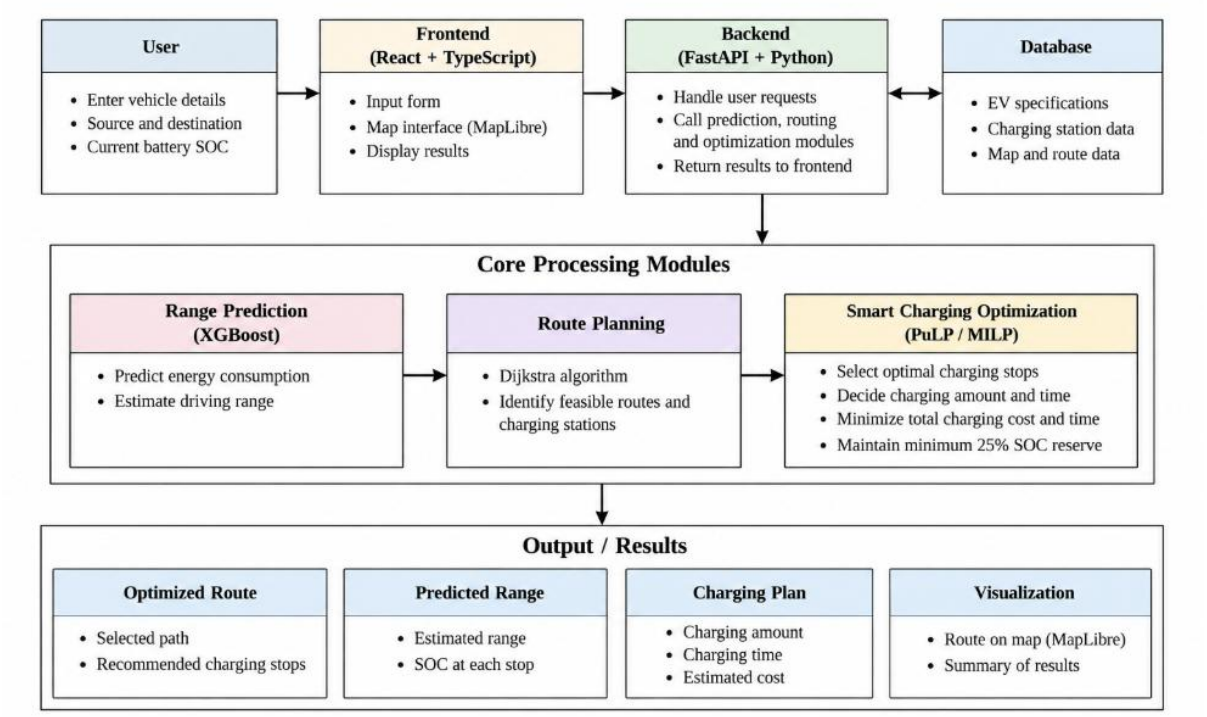


Fig. 3.1: System Architecture of the Proposed EV Range Prediction and Smart Charging Optimization System

3.3 DESCRIPTION OF MODULES

The proposed system contains five modules that work together to produce the final trip plan.

Module 1: Energy Prediction

This module estimates trip energy using XGBoost. The input features describe the EV, battery, route and operating conditions. The model returns a continuous energy value in kWh, which is passed to the battery-estimation stage.

Module 2: Range and SOC Estimation

This module estimates remaining battery energy and arrival SOC. The available energy is calculated from battery capacity and initial SOC. Predicted trip energy is subtracted to estimate remaining energy. The result is converted to an SOC percentage, and the safe range is estimated using the configured 25% reserve.

Module 3: Charging Station Selection

This module is activated when the estimated arrival SOC falls below the reserve. It identifies stations related to the selected route and checks whether they satisfy the modeled reachability requirements. Only eligible candidates are passed to the optimization stage.

Module 4: Smart Charging Optimization

This module formulates a MILP model using PuLP. A binary decision variable represents whether a candidate station is selected, while a continuous variable represents the amount of energy to charge. The objective considers estimated charging cost and time while respecting the encoded constraints.

Module 5: Trip Monitoring and Dashboard

This module displays the selected route, distance, travel duration, predicted energy, arrival SOC, safe range and charging status. When a feasible charging plan is returned, it also shows the recommended station and estimated charging amount, time and cost.

The modules separate prediction, battery calculations and charging decisions, making the system easier to explain and test. The dashboard brings the results together so that users can understand the complete trip-planning outcome.

3.4 DATA COLLECTION & DATA PREPROCESSING

The energy-prediction dataset contains 20,000 structured EV trip records. The records include features describing vehicle characteristics, battery information, trip distance, driving conditions and environmental factors. The target variable is trip energy consumption measured in kWh.

The input feature groups include manufacturer, vehicle model, battery capacity, battery type, motor power, vehicle weight, route distance, average speed, road type, temperature, weather, state of health and driving style. These variables provide the model with information that may be associated with trip-energy requirements.

Before training, the data must be checked for missing values, duplicate records, invalid numeric values and inconsistent units. Categorical fields are converted into a numeric representation using one-hot encoding. Numeric features are prepared in a consistent form for model training and inference.

A vehicle-aware split is used to reduce the risk of the same vehicle appearing in both the training and test partitions. The reported split contains 16,050 training records and 3,950 test records. The training partition contains 4,299 vehicles, while the test partition contains 1,075 vehicles.

Five-fold GroupKFold cross-validation is used in the reported model comparison. Grouping by vehicle helps evaluate performance on vehicle groups excluded from the corresponding training fold. This approach reduces a potential source of leakage, although it does not replace independent validation using actual EV telemetry.

The dataset target values are used as the ground truth for model evaluation. They should not be described as measurements collected from a physical EV during this project unless separate sensor measurements are available.

Dataset summary

Item	Description
Total records	20,000
Training records	16,050
Testing records	3,950
Training vehicles	4,299
Testing vehicles	1,075
Target variable	Trip energy consumption
Target unit	kWh
Validation method	Five-fold GroupKFold

3.5 ENERGY PREDICTION MODEL

XGBoost is used to predict trip energy consumption. It is a gradient-boosted decision-tree method that builds an ensemble of trees to improve predictions over successive boosting stages. It is suitable for structured tabular data and can model nonlinear relationships between vehicle features, trip distance and operating conditions.

The model receives the processed input features and predicts a continuous value representing the energy required for the selected trip. This makes the task a supervised regression problem. The prediction is reported in kWh and is used by the SOC/range module.

The final XGBoost configuration uses 300 estimators, a maximum tree depth of 6, a learning rate of 0.05, row subsampling of 0.9 and feature subsampling of 0.9. The objective is reg:squarederror, the evaluation metric is RMSE, and the random state is 42.

Five regression models were evaluated: Random Forest, Extra Trees, Gradient Boosting, XGBoost and CatBoost. XGBoost achieved the best reported test metrics in the comparison, with R^2 = 0.9978, MAE = 0.1481 and RMSE = 0.3613

The R^2 value indicates a strong fit to the target variation in the held-out test data. It should not be described as 99.78% prediction accuracy. MAE represents the average absolute prediction error, while RMSE gives greater weight to larger errors.

Final XGBoost parameters

Parameter	Value
Number of estimators	300
Maximum depth	6
Learning rate	0.05
Subsample	0.9
Column subsample	0.9
Objective	reg
Evaluation metric	RMSE
Random state	42

3.6 PROPOSED METHODOLOGY

The proposed methodology begins when the user enters an origin, destination, EV model and initial SOC. The system validates the input and resolves the geographic locations into coordinates. These coordinates are matched to the available road graph.

The routing component generates a primary route using Dijkstra's algorithm and provides alternative paths for user selection. The selected route's distance and duration are returned to the interface. The selected path is then used to prepare the energy-prediction inputs.

XGBoost predicts the trip energy. The SOC/range module calculates the available energy at departure using battery capacity and initial SOC, subtracts the predicted trip energy and estimates the remaining battery state. The system compares estimated arrival SOC with the 25% planning reserve.

If arrival SOC is at or above the reserve, the trip is reported as feasible without charging under the model assumptions. If arrival SOC is below the reserve, the system identifies candidate charging stations and filters them according to route relevance and reachability.

The feasible candidates are passed to the PuLP/MILP optimization stage. The optimizer evaluates station selection and charging quantity based on the configured cost/time objective and constraints.

If a valid solution exists, the recommended station and estimated charging details are returned to the dashboard. If no feasible solution exists, the system reports that result explicitly.

The final dashboard displays the route, predicted energy, estimated SOC, safe range and charging decision. The workflow combines geographic processing, machine-learning prediction and optimization into a single trip-planning process.

Proposed methodology flow

1. Enter origin, destination and EV parameters.
2. Resolve locations and map them to the road graph.
3. Generate primary and alternative routes.
4. Select the preferred route.
5. Predict trip energy using XGBoost.
6. Estimate arrival SOC and safe range.
7. Check whether arrival SOC satisfies the 25% reserve.
8. If feasible, display the trip plan without charging.
9. If not feasible, filter reachable charging stations.
10. Run the PuLP/MILP charging optimizer.
11. Display a feasible charging recommendation or a no-feasible-station result.

3.7 SUMMARY

This chapter described the system requirements, architecture, five functional modules, dataset preparation and energy-prediction methodology. The system uses a React/TypeScript interface, FastAPI backend, geographic routing tools, XGBoost regression and PuLP-based charging optimization.

The vehicle-aware dataset split supports a more independent evaluation of the energy model, while the SOC reserve check controls the charging workflow. The next chapter describes how these components are implemented and connected in the application.

CHAPTER 4

SYSTEM IMPLEMENTATION

4.1 GENERAL

The system implementation converts the proposed methodology into a working web-based EV trip-planning application. It connects the frontend, backend, geographic routing, energy-prediction model, battery calculations and charging optimizer into one workflow.

The frontend is developed using React and TypeScript. MapLibre GL JS provides map visualization, while FastAPI and Python handle backend processing. OSMnx and NetworkX support road-network processing, XGBoost performs energy prediction, PuLP formulates the charging optimization problem, and MongoDB stores project and geographic data.

The implementation is divided into five modules: Energy Prediction, Range and SOC Estimation, Charging Station Selection, Smart Charging Optimization, and Trip Monitoring and Dashboard. Each module receives input from the preceding stage and produces output for the next stage.

The system is intended for trip-planning decision support. It estimates energy requirements and charging needs based on the selected route, user inputs, dataset and configured constraints. It does not directly control the EV or replace the vehicle's battery-management system.

4.2 SYSTEM WORKFLOW

The workflow begins when the user enters the origin and destination and selects the EV model and initial SOC. The application validates the input and resolves the locations to geographic coordinates. These coordinates are then connected to the road network used for route generation.

The route engine generates a primary route and alternative routes. The user selects the preferred route, and its distance and duration are used in the subsequent energy-prediction process. The backend prepares the feature vector and passes it to the trained XGBoost model.

The model predicts trip energy in kWh. The SOC/range module uses this prediction, initial SOC and battery capacity to estimate arrival SOC and safe range. The estimated arrival SOC is checked against the minimum 25% planning reserve.

If the estimated arrival SOC satisfies the reserve, the system displays the trip as feasible without charging under the model assumptions. If it falls below the reserve, the system searches for charging-station candidates and checks reachability.

Feasible candidates are passed to the PuLP/MILP optimizer. The optimizer determines a charging recommendation when a valid solution exists. The final dashboard displays the route, energy prediction, SOC estimate, safe range and charging result. If no feasible station satisfies the constraints, the application reports that outcome.

SYSTEM WORKFLOW:

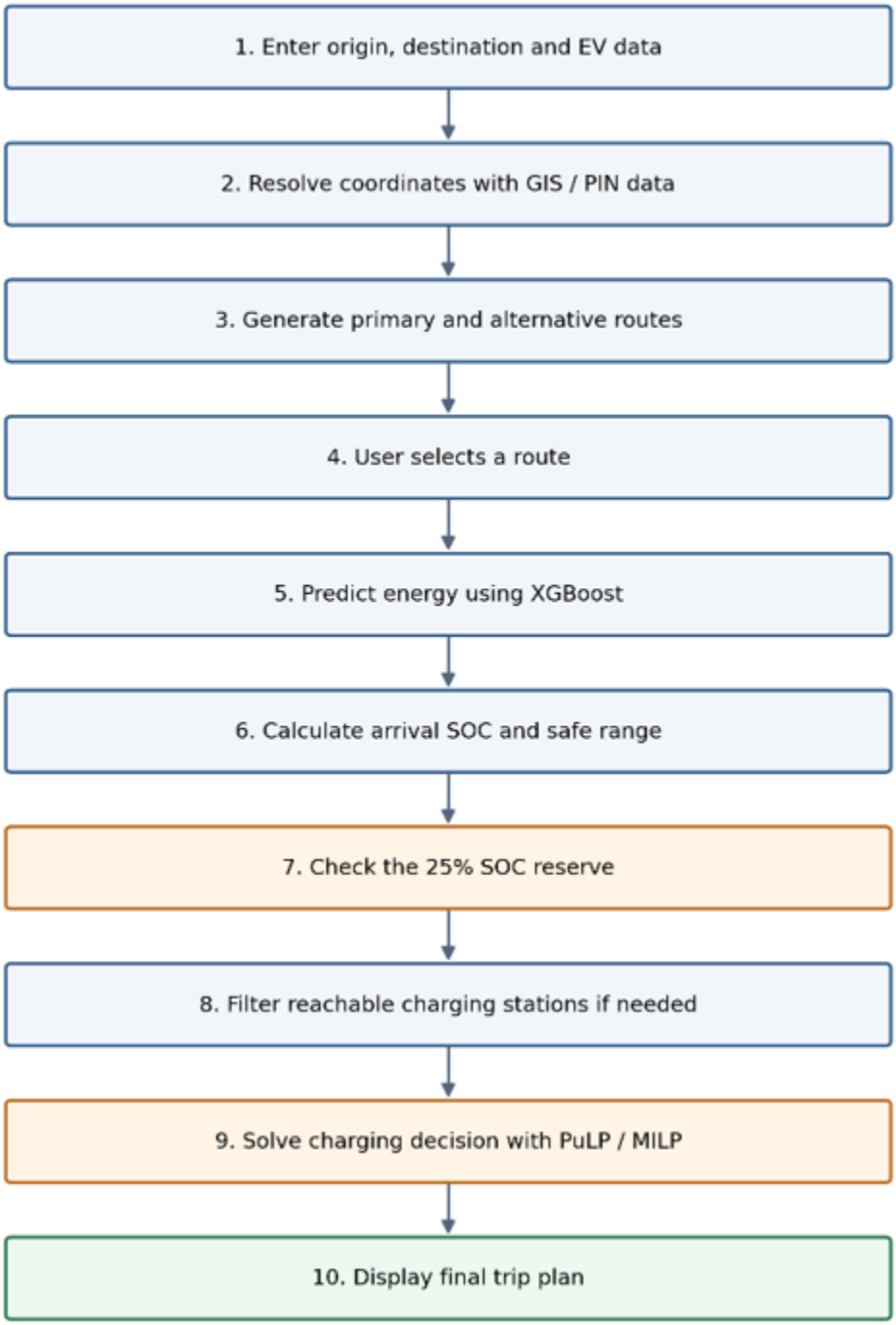


Figure 4.1. End-to-end workflow for location selection, route planning, prediction and charging.

4.3 OVERVIEW OF THE PLATFORM

The platform provides a map-based interface for planning EV trips. Users can provide an origin and destination using the supported location-entry options, select an EV model and enter the starting SOC and relevant operating parameters.

The map displays the selected locations and route options. Route information allows users to compare distance and estimated duration before choosing a path. Once a route is selected, the backend performs energy prediction and battery-state calculations.

The results panel displays the predicted energy consumption, estimated arrival SOC, safe range and charging status. When charging is required and a feasible solution is available, the platform displays the recommended charging station and the estimated charging quantity, duration and cost.

The frontend uses React and TypeScript, with MapLibre GL JS for mapping. The backend uses FastAPI and Python to coordinate the processing stages. Geographic routing is handled by OSMnx and NetworkX, and MongoDB supports project data storage.

The platform combines information from several modules into a single interface so that the user can understand the overall trip-planning result. Its outputs are estimates based on the model, route data and configured assumptions.

4.4 MODULE IMPLEMENTATION

Module 1 – Energy Prediction

The energy-prediction module loads the trained XGBoost regression model and prepares the selected EV and route information as input features. The same preprocessing and feature representation used during training must be applied during inference.

The model returns a predicted trip-energy value in kWh. This prediction is passed to the SOC/range module. The model does not directly measure battery state or predict SOC as its target.

Module 2 – Range and SOC Estimation

The SOC module calculates available battery energy from battery capacity and initial SOC. Predicted trip energy is subtracted to estimate remaining energy. The result is converted into an estimated arrival SOC.

The module also estimates safe range using the available energy above the 25% reserve. These values are planning estimates and may differ from actual driving results because of real-world traffic, weather, driving behaviour and battery conditions.

Module 3 – Charging Station Selection

When the estimated arrival SOC falls below the reserve, the station-selection module identifies candidate stations associated with the selected route. It checks whether each candidate satisfies the implemented reachability conditions.

Stations that fail the modeled conditions are excluded before optimization. The remaining candidates are passed to the charging optimizer. The system does not assume that the nearest station is automatically the best choice.

Module 4 – Smart Charging Optimization

The charging-optimization module uses PuLP to formulate a Mixed-Integer Linear Programming model. A binary variable represents whether a station is selected, while a continuous variable represents the charging quantity.

The objective considers the configured charging cost and charging time. The model also applies the constraints implemented for station selection, battery capacity and minimum SOC reserve. If a feasible solution exists, the system returns the selected station and charging amount. Otherwise, it reports the infeasible result.

Charging cost and time are estimates whenever verified station-specific power and tariff data are unavailable. The application provides recommendations; it does not automatically initiate charging.

Module 5 – Trip Monitoring and Dashboard

The dashboard displays the selected route, route distance, travel duration, predicted energy, estimated arrival SOC, safe range and charging status. It presents charging details when a feasible recommendation is returned.

The dashboard also makes the result easier to interpret by connecting the route and battery estimates with the final decision. It is a software interface for trip planning rather than a direct vehicle telemetry display.

4.5 SUMMARY

This chapter described the implementation and workflow of the EV trip-planning application. The platform connects route selection, XGBoost energy prediction, SOC/range estimation, charging-station filtering and PuLP/MILP optimization.

The modular implementation allows each stage to be tested separately and then evaluated as part of the complete workflow. The next chapter presents the experimental setup, model metrics, route results and charging scenarios.

CHAPTER 5

EXPERIMENTAL ANALYSIS

5.1 EXPERIMENTAL SETUP

The experiment evaluates the energy-prediction model and the integrated trip-planning workflow. The prediction dataset contains 20,000 structured EV trip records. The vehicle-aware split uses 16,050 records for training and 3,950 records for testing. Vehicle identities are separated between the two partitions to reduce leakage from trips associated with the same vehicle.

Five regression models are compared: Random Forest, Extra Trees, Gradient Boosting, XGBoost and CatBoost. Model performance is evaluated using R^2, Mean Absolute Error (MAE) and Root Mean Square Error (RMSE). Five-fold GroupKFold cross-validation results are also reported for model comparison.

The system-level tests examine route generation, energy prediction, SOC estimation, the 25% reserve decision and charging optimization. Representative scenarios are used to test trips that do not require charging, trips that activate the charging process and cases where no feasible charging station is returned.

The end-to-end test uses a BYD Atto 3 with an initial SOC of 80%, a route distance of 14.77 km and an estimated duration of 22 minutes. The model predicts 2.55 kWh of energy, and the estimated arrival SOC is 75.8%.

The experiments evaluate the current prototype using the available dataset and configured assumptions. The test metrics do not establish independent real-world fleet validation.

5.2 PERFORMANCE EVALUATION

The performance of the energy-prediction model is evaluated using three regression metrics.

Mean Absolute Error (MAE) measures the average absolute difference between predicted and target energy. It is expressed in kWh and provides a direct measure of average prediction error.

Root Mean Square Error (RMSE) is calculated by squaring the residuals, taking their mean and then calculating the square root. Larger errors have a greater influence on RMSE than on MAE.

R^2, the coefficient of determination, describes how much of the target variation is explained by the predictions relative to the mean-target baseline. It is not a classification accuracy percentage.

Performance Evaluation Metrics

The performance of the energy-prediction model is evaluated using Mean Absolute Error (MAE), Root Mean Square Error (RMSE) and the coefficient of determination (R^2). These metrics measure the difference between the predicted energy values and the corresponding target values in the held-out test dataset.

Mean Absolute Error (MAE)

MAE measures the average absolute difference between the target energy values and the predicted energy values. A lower MAE indicates that the predictions are closer to the target values on average.

$$MAE = \frac{1}{n}\sum_{i=1}^{n} | y_i - \hat{y}_i |$$

Root Mean Square Error (RMSE)

RMSE is calculated by taking the square root of the mean of the squared prediction errors. Since the errors are squared, larger deviations have a greater influence on this metric.

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}$$

Coefficient of Determination (R^2)

The coefficient of determination measures how well the model predictions explain the variation in the target energy values compared with using their mean as a baseline.

$$R^2 = 1 - \frac{\sum(y_i - \hat{y}_i)^2}{\sum(y_i - \bar{y})^2}$$

Actual Versus Predicted Energy Consumption:

The actual-versus-predicted graph evaluates the performance of the XGBoost regression model by comparing the target energy values in the test dataset with the corresponding energy values predicted by the model. The horizontal axis represents actual energy consumption in kWh, while the vertical axis represents predicted energy consumption in kWh.

The diagonal line represents ideal prediction, where the predicted value is equal to the target value. Data points located close to this line indicate that the model predictions are close to the corresponding target values.

In the plotted results, most data points are concentrated near the diagonal line, indicating a strong agreement between the predicted and target energy values for most test records.

The model achieved a test R^2 of 0.9978, an MAE of 0.1481 and an RMSE of 0.3613 across 3,950 test records. These metrics support the use of XGBoost for energy prediction in the proposed EV trip-planning system. However, some deviations from the ideal line are visible, particularly for certain higher-energy records. Therefore, the graph indicates strong predictive performance on the available test dataset rather than perfect prediction under every driving condition.

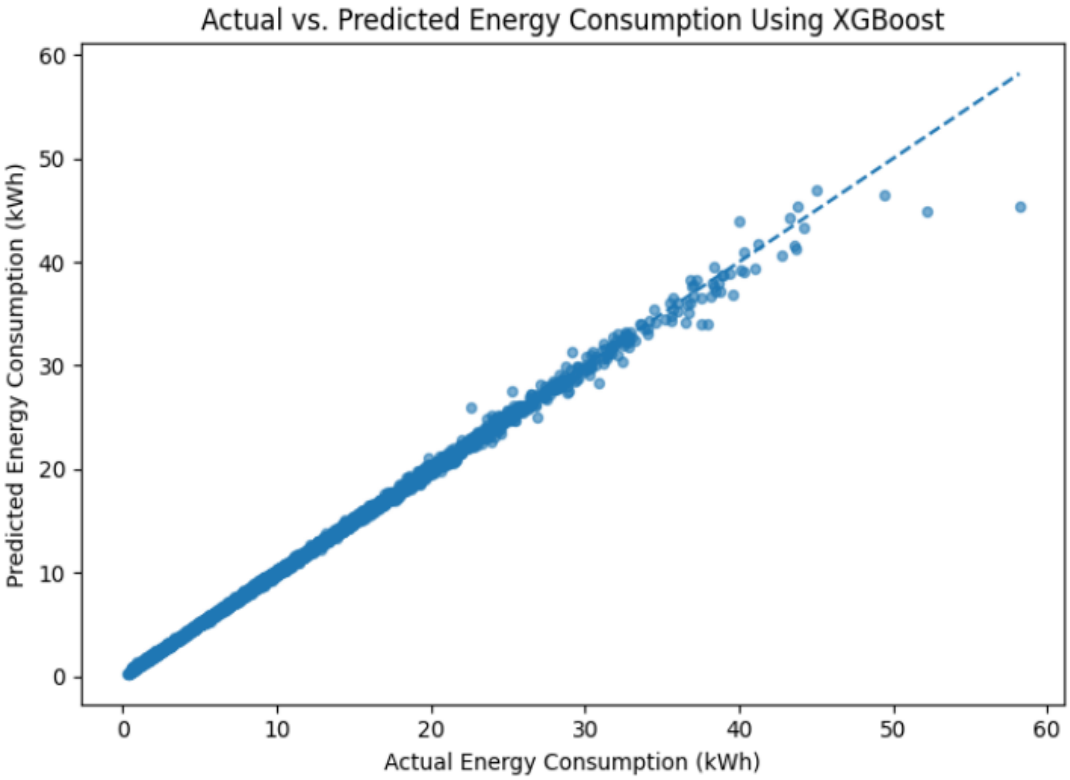


Figure 5.1. Actual versus predicted energy on the held-out test dataset.

Residual Error Analysis:

Residual analysis is used to examine the differences between the target energy values and the energy values predicted by the XGBoost model. A residual is calculated by subtracting the predicted value from the corresponding target value. In the residual error plot, the horizontal axis represents predicted energy consumption in kWh, and the vertical axis represents residual error in kWh. The horizontal reference line at zero indicates predictions for which the target and predicted values are equal.

Most residuals are distributed around the zero-error line, suggesting that the model predictions are close to the target values for a large proportion of the test records. However, a small number of points show larger positive or negative residuals, particularly at higher predicted energy values. Positive residuals indicate that the model predicted less energy than the target value, whereas negative residuals indicate that the model predicted more energy than the target value.

The presence of these larger residuals shows that the model does not produce identical prediction errors for every trip. This observation is consistent with the RMSE of 0.3613 kWh being higher than the MAE of 0.1481 kWh, since RMSE gives greater weight to larger errors. The residual plot therefore provides additional information about prediction errors that cannot be understood from a single performance metric alone.

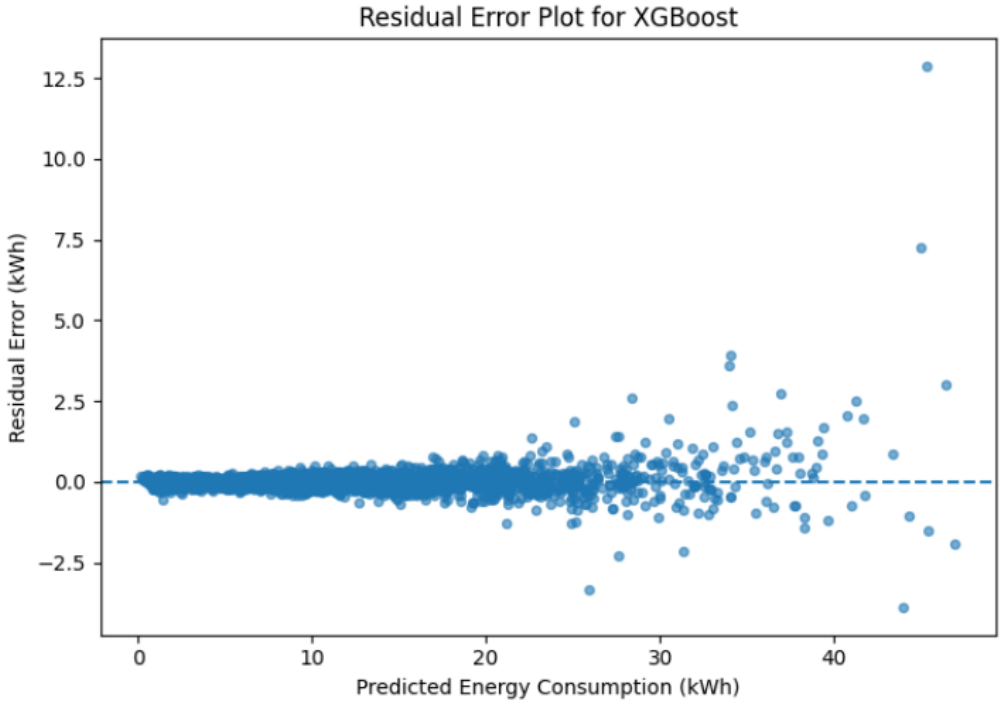


Figure 5.2. Residual error distribution for the XGBoost energy model.

Discussion of Prediction Results:

The model comparison table and the two graphical evaluations provide complementary evidence of the XGBoost model's performance. The model comparison identifies XGBoost as the best-performing model among the five evaluated algorithms according to the reported test metrics. The actual-versus-predicted graph visually demonstrates the agreement between the model outputs and the target values, while the residual plot highlights the magnitude and direction of the prediction errors.

Together, these results support the selection of XGBoost as the energy-prediction component of the proposed system. The predicted trip energy is subsequently used to estimate arrival State of Charge (SOC), calculate the remaining safe range and determine whether charging analysis is required. Nevertheless, the results are limited to the available structured dataset and the implemented experimental setup. Independent evaluation using measured real-world EV trip data is required to establish performance under actual operating conditions.

5.3 COMPARISON WITH EXISTING WORK

The literature reviewed in this project covers EV energy prediction, graph-based routing, charging-station recommendation and optimization. Some studies focus on detailed powertrain models, while others use machine learning or ensemble techniques to estimate energy consumption. Research also examines how predicted energy can be combined with route planning and charging decisions.

The proposed system uses XGBoost because it performed strongly in the project's model comparison and supports regression on structured tabular features. Dijkstra-based routing provides the primary path, while alternative routes allow the user to choose a route before the energy prediction is performed. PuLP/MILP then supports reserve-constrained charging decisions.

The main difference is the integration of these functions into a single web-based trip-planning application. The system combines location resolution, route selection, energy prediction, SOC estimation and charging optimization rather than presenting each as a separate tool.

The comparison should not be interpreted as proof that the proposed system outperforms every published method. Different studies may use different datasets, features, evaluation splits and prediction targets. The reported XGBoost metrics are directly comparable only with results evaluated under sufficiently similar conditions.

The contribution of this project is the integrated implementation and evaluation of a trip-planning workflow using the available dataset, geographic information and optimization model.

5.4 RESULT & DISCUSSION

The model comparison shows that XGBoost achieved the strongest reported test metrics among the five evaluated models. Its test R^2 is 0.9978, MAE is 0.1481 and RMSE is 0.3613, These results support using XGBoost as the energy-prediction component in the current prototype.

The route evaluation demonstrates that the system can present a primary route and alternative options. In the reported example, the primary route is 20.05 km with a duration of 25 minutes, Alternative 1 is 23.15 km with a duration of 34 minutes, and Alternative 2 is 28.18 km with a duration of 47 minutes. These figures represent a specific route test, not general travel-time benchmarks for Chennai.

The charging tests demonstrate the effect of the reserve constraint. In Scenario S1, initial SOC is 28%, predicted energy is 3.51 kWh and estimated arrival SOC is 22.2%. The charging workflow is activated, and a feasible station recommendation is returned in the reported scenario.

In Scenario S2, initial SOC is 20%, predicted energy is 3.51 kWh and arrival SOC is 14.2%. In Scenario S3, initial SOC is 20%, predicted energy is 3.84 kWh and arrival SOC is 13.7%. No feasible station is found under the modeled candidate set and constraints for either scenario. The system reports this result instead of forcing a recommendation.

The end-to-end test uses a BYD Atto 3 with an initial SOC of 80%. The route distance is 14.77km, the estimated duration is 22 minutes and predicted energy is 2.55 kWh. The safe-range estimate is 192.9 km and arrival SOC is 75.8%, so charging is not required under the planning assumptions.

These results demonstrate the connection between route planning, prediction, battery estimation and charging decisions. However, the energy and SOC values remain estimates, and charging power or price may rely on prototype assumptions. Further validation with actual vehicle telemetry and verified charging-station information is needed before real-world deployment.

Table 5.1 – Model Comparison

Model	Test R^2	MAE	RMSE
Random Forest	0.9947	0.3019	0.5605
Extra Trees	0.9975	0.1930	0.3838
Gradient Boosting	0.9970	0.2649	0.4243
XGBoost	**0.9978**	**0.1481**	**0.3613**
CatBoost	0.9972	0.1744	0.4044

The performance of five machine learning regression models was evaluated using the held-out test dataset. The models considered were Random Forest, Extra Trees, Gradient Boosting, XGBoost and CatBoost. Their performance was compared using the coefficient of determination (R^2), Mean Absolute Error (MAE), Root Mean Square Error (RMSE) and cross-validation results.

Among the evaluated models, XGBoost achieved the highest test R^2 of 0.9978, the lowest MAE of 0.1481 and the lowest RMSE of 0.3613, These results indicate that XGBoost provided the strongest predictive performance among the five models under the experimental setup used in this project. Therefore, XGBoost was selected as the energy-prediction model for the integrated EV trip-planning application.

The reported results are based on 3,950 held-out test records from the structured EV trip dataset. They describe agreement between model predictions and the dataset's target energy values and should not be interpreted as independent validation using measurements collected from a real-world EV fleet.

Table 5.2 – Charging Optimization Scenarios

Scenario	Initial SOC	Predicted Energy	Arrival SOC	Result
S1	28%	3.51 kWh	22.2%	Feasible charging recommendation
S2	20%	3.51 kWh	14.2%	No feasible station found
S3	20%	3.84 kWh	13.7%	No feasible station found

Table 5.3 – End-to-End Test

Parameter	Result
EV model	BYD Atto 3
Initial SOC	80%
Route distance	14.77 km
Duration	22 minutes
Predicted energy	2.55 kWh
Estimated safe range	192.9 km
Estimated arrival SOC	75.8%
Charging required	No

5.5 SYSTEM-LEVEL TEST CASES

System-level testing is performed to evaluate the integrated operation of the AI-Assisted EV Range Prediction and Smart Charging Optimization System. The tests examine route generation, energy prediction, battery SOC estimation, reserve checking and charging-station optimization. The purpose is to verify how the different modules work together and how the application responds to different trip conditions.

The evaluation uses the documented route results, held-out machine-learning test data and representative EV trip scenarios. The energy values and battery states are estimates produced by the system rather than independent measurements collected from a physical EV fleet.

Table 5.4: System-Level Test Cases

Test ID	Test Case	Input / Condition	Observed Result
TC-01	Route generation	Origin, destination and request for alternative routes	Primary route and alternative routes are returned.
TC-02	Energy prediction	3,950 held-out test records	XGBoost achieves $R^2 = 0.9978$, MAE = 0.1481 kWh and RMSE = 0.3613
TC-03	Trip feasible without charging	BYD Atto 3, initial SOC 80%, route distance 14.77 km	Predicted energy is 2.55 kWh; estimated arrival SOC is 75.8%; charging is not required.
TC-04	Charging required with a feasible option	Scenario S1: initial SOC 28%, predicted energy 3.51 kWh	Arrival SOC is estimated as 22.2%; the system returns a feasible charging recommendation.
TC-05	No feasible charging station	Scenario S2: initial SOC 20%, predicted energy 3.51 kWh	Arrival SOC is estimated as 14.2%; no feasible station is returned.
TC-06	No feasible charging station	Scenario S3: initial SOC 20%, predicted energy 3.84 kWh	Arrival SOC is estimated as 13.7%; no feasible station is returned.

5.5.1 Route Generation Testing

The route-generation test evaluates whether the application can produce a primary route and alternative routes for a selected origin and destination. In the documented example, the primary route is 20.05 km with an estimated duration of 25 minutes. Alternative Route 1 is 23.15 km with an estimated duration of 34 minutes, while Alternative Route 2 is 28.18 km with an estimated duration of 47 minutes.

The application returns the available route options for user selection. The selected route is then used by the energy-prediction module. This test demonstrates how route generation and energy prediction are connected within the trip-planning workflow.

5.5.2 Energy Prediction Testing

The XGBoost model is evaluated using 3,950 held-out test records. It achieves an R^2 of 0.9978, MAE of 0.1481 kWh and RMSE of 0.3613 kWh. These metrics indicate close agreement between the predicted energy values and the target values in the structured dataset.

The actual-versus-predicted graph and residual error plot provide additional information about the model's behaviour. Most points are close to the ideal-prediction line, while some larger errors occur at higher energy values. The results support the selection of XGBoost for the current prototype, but they do not establish independent real-world fleet validation.

5.5.3 Battery Reserve and Charging Tests

The battery reserve test checks whether the predicted arrival SOC remains at or above the configured 25% threshold. The end-to-end test begins with an initial SOC of 80%, and the estimated arrival SOC is 75.8%. Therefore, charging is not required under the model assumptions.

Scenario S1 begins with an SOC of 28%. The predicted journey energy is 3.51 kWh, resulting in an estimated arrival SOC of 22.2%. Because this value is below the 25% reserve, the system activates charging optimization and returns a feasible recommendation.

Scenarios S2 and S3 begin with an SOC of 20%. Their estimated arrival SOC values are 14.2% and 13.7%, respectively. Under the current candidate-station set and strict reserve constraints, no feasible charging station is returned. This demonstrates that the system reports an infeasible outcome instead of automatically selecting a station that violates its planning conditions.

Overall, the documented test cases demonstrate the main application workflow, including route selection, energy prediction, SOC estimation, charging recommendation and infeasibility handling.

5.6 DETAILED ANALYSIS OF CHARGING SCENARIOS

The charging-optimization module is activated when the predicted trip energy indicates that the vehicle cannot complete the selected journey while maintaining the configured minimum SOC reserve of 25%. The module evaluates charging-station candidates and attempts to identify a feasible charging decision based on the modeled energy, station reachability, battery capacity, charging cost and charging time.

Three scenarios, S1, S2 and S3, are used to examine the effect of initial SOC and predicted trip energy on the charging decision.

Table 5.5: Detailed Charging Scenario Comparison

Parameter	Scenario S1	Scenario S2	Scenario S3
Initial SOC	28%	20%	20%
Predicted trip energy	3.51 kWh	3.51 kWh	3.84 kWh
Estimated arrival SOC	22.2%	14.2%	13.7%
Minimum SOC reserve	25%	25%	25%
Charging analysis	Activated	Activated	Activated
Optimization outcome	Feasible charging recommendation	No feasible station found	No feasible station found

5.6.1 Scenario S1 — Charging Recommendation Available

Scenario S1 begins with an initial SOC of 28%. The XGBoost model predicts that the selected journey requires 3.51 kWh of energy. After this predicted consumption is taken into account, the estimated arrival SOC is 22.2%.

Since the estimated arrival SOC is below the minimum reserve of 25%, the application identifies the need for intermediate charging. The charging-station selection module supplies eligible candidates to the PuLP-based MILP optimizer. In this documented scenario, a feasible charging recommendation is returned.

This scenario demonstrates the connection between energy prediction, reserve checking and charging optimization. Rather than relying only on the distance to the destination, the system uses predicted trip energy to determine whether charging is needed.

The charging recommendation remains dependent on the candidate stations and constraints available to the prototype. It should therefore be interpreted as an optimization result under the implemented assumptions, not a guarantee that a station will be available in real time.

5.6.2 Scenario S2 — No Feasible Station Found

Scenario S2 starts with an initial SOC of 20%, and the predicted trip energy is 3.51 kWh. The estimated arrival SOC is 14.2%, which is below the configured minimum reserve.

In this scenario, the initial SOC is already lower than the 25% planning reserve before the journey begins. Consequently, the charging-station selection process must consider whether a candidate can be reached without violating the implemented reserve condition. The system does not find a candidate that satisfies the conditions in the evaluated scenario.

The result indicates that charging is required but that the current model cannot recommend a feasible station. This is an important behaviour because selecting a station simply because it is nearby could produce a recommendation inconsistent with the reserve constraint.

The system therefore reports that no feasible station was found rather than forcing an invalid charging decision. In practical use, such a result indicates that the current route, initial battery state or available station candidates require reconsideration.

5.6.3 Scenario S3 — Higher Predicted Energy Requirement

Scenario S3 also begins with an initial SOC of 20%, but the predicted trip energy increases to 3.84 kWh. The estimated arrival SOC is 13.7%, which is lower than the result obtained in S2.

The lower estimated arrival SOC reflects the greater predicted energy requirement under the stated scenario. Since the initial SOC is below the 25% reserve and the available candidate set does not provide a feasible recommendation, the optimizer returns no feasible station.

Comparison with S2 shows that the charging decision depends on both initial battery state and predicted energy consumption. A greater predicted energy requirement can make trip feasibility more restrictive. The result also demonstrates why the application needs to check station reachability and battery constraints before producing a charging recommendation.

5.6.4 Comparative Discussion

The three scenarios demonstrate two different outcomes of the charging workflow. S1 produces a feasible charging recommendation, while S2 and S3 produce an explicit no-feasible-station result. These outcomes show how the application responds to different initial SOC and predicted energy conditions.

The 25% reserve acts as a planning constraint in the current implementation. The system uses it to assess whether the trip can be completed under the available energy estimate and whether a charging candidate satisfies the implemented conditions. It does not replace the vehicle's Battery Management System or provide a universal safety guarantee.

The charging-optimization process also depends on the available geographic and station data. Where verified station availability, power ratings and tariffs are unavailable, charging time and cost may rely on the assumptions used by the prototype. Consequently, these results demonstrate the decision-making behaviour of the implemented model rather than confirmed real-time charging availability.

In conclusion, the scenario analysis shows that the proposed system can recommend charging when a feasible option exists and explicitly report infeasibility when the evaluated candidates do not satisfy the constraints. This approach supports more transparent trip-planning decisions and avoids presenting an unsupported charging recommendation as a valid result.

5.7 SUMMARY

This chapter evaluated the XGBoost prediction model and the integrated trip-planning workflow. XGBoost achieved the best reported test metrics among the evaluated regression models, while the route and charging scenarios demonstrated how predicted energy is used to estimate arrival SOC and determine whether charging is required.

The system can return a feasible charging recommendation when the modeled conditions permit it and report infeasibility when no candidate satisfies the constraints. The results support the implementation as a decision-support prototype, while further real-world validation remains necessary.

CHAPTER 6

CONCLUSION AND FUTURE WORK

6.1 CONCLUSION

The project developed an AI-Assisted EV Range Prediction and Smart Charging Optimization System that integrates route planning, energy prediction, SOC/range estimation and charging optimization in a web-based application.

The system accepts trip locations and EV parameters, generates route options, predicts energy consumption using XGBoost and estimates arrival SOC and safe range. A minimum 25% SOC reserve controls whether charging analysis is required. When necessary, charging-station candidates are filtered and evaluated using a PuLP-based MILP model.

The application uses React and TypeScript for the frontend, MapLibre GL JS for map visualization, FastAPI and Python for backend processing, OSMnx and NetworkX for routing, MongoDB for data storage, XGBoost for prediction and PuLP for optimization.

XGBoost achieved a test R^2 of 0.9978, MAE of 0.1481 and RMSE of 0.3613 on the available structured dataset. The end-to-end test demonstrated a trip with no charging required, while the low-SOC scenarios exercised the charging recommendation and infeasibility branches.

The project demonstrates how machine learning and mathematical optimization can be integrated into EV trip planning. Its results are limited to the available dataset and modeled assumptions and do not establish independent real-world fleet validation.

6.2 LIMITATION OF CURRENT WORK

The energy-prediction model was evaluated using structured EV trip records rather than a large independently collected fleet-telemetry dataset. The reported metrics therefore describe performance on the available data and may not represent prediction errors under all real-world driving conditions.

The route-planning process depends on the available road graph and its edge weights. Live traffic conditions are not fully incorporated into the current routing objective. Energy predictions may also be less reliable for vehicles, route types or operating conditions that are underrepresented in the training data.

The SOC and range outputs are calculated estimates. The system does not directly measure the physical battery state or replace the vehicle's BMS. The 25% SOC reserve is a configurable planning rule in the current implementation, not a universal safety guarantee.

Charging cost, duration and station suitability depend on the information available to the prototype. Where verified live station power, availability and tariffs are unavailable, the system uses modeled or assumed values. The application does not automatically initiate charging or guarantee that a recommended station is available at the time of travel.

6.3 FUTURE ENHANCEMENT

Future work can improve prediction reliability by evaluating the model against independently collected EV telemetry and measured trip-energy data. A larger dataset covering more vehicle models, battery conditions, route types, traffic patterns and weather conditions would help assess generalization.

The routing component can be extended to use live traffic data and include predicted energy consumption in route selection. This would allow route alternatives to be compared using travel time, distance, energy requirements and charging implications rather than relying mainly on the current route-weighting strategy.

Charging optimization can be improved by integrating verified station locations, connector compatibility, live availability, charging power and tariffs. This would provide more realistic charging-time and cost estimates. The optimization objective could also be extended to support user preferences such as minimizing detour distance, charging cost or total journey time.

Other possible enhancements include configurable reserve thresholds, more robust predictions for long-distance journeys, additional geographic coverage and live vehicle-location integration. These changes should be tested against appropriate data and operational requirements before the system is used for real-world trip decisions.

6.4 SUMMARY

This chapter concluded that the project successfully integrates energy prediction, route planning, SOC/range estimation and charging optimization into one application. The reported model metrics and trip scenarios demonstrate the operation of the current prototype.

The main limitations concern independent real-world validation, live traffic and charger data, and the use of estimated rather than measured battery values. Future improvements should focus on stronger real-world datasets, verified charging information and more comprehensive route-energy optimization.

CHAPTER 7

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[9] Greater Chennai Corporation, geographic information system records used for Chennai location resolution. The final submission should identify the actual source layer and acquisition date.

[10] PuLP Project, “PuLP Documentation.” Python library used to formulate linear and mixed-integer optimization models.

[11] OSMnx Developers, “OSMnx Documentation.” Python library used for street-network acquisition and analysis.

[12] NetworkX Developers, “NetworkX Documentation.” Python library used for graph representation and path algorithms.

[13] XGBoost Developers, “XGBoost Documentation.” Documentation for the regression estimator and model implementation.

[14] FastAPI, “FastAPI Documentation.” Framework used to develop the backend API.

[15] MapLibre, “MapLibre GL JS Documentation.” Mapping library used in the frontend.

[16] MongoDB, “MongoDB Documentation.” Database and geospatial query documentation relevant to project storage.

CHAPTER 8

SOURCE CODE

This chapter presents the main application code for the AI-Assisted EV Range Prediction and Smart Charging Optimization System. It includes the frontend, backend API, route generation, energy prediction, SOC estimation and charging optimization.

8.1 FRONTEND APPLICATION

Source file: frontend/src/App.tsx

The frontend sends route-planning and EV-analysis requests to the backend and displays the returned results.

Listing 8.1: Route Planning and EV Analysis

```
const API = "http://127.0.0.1:8000/api";
const planRoute = async () => {
  if (!start || !destination) {
    setError("Set both start and destination first.");
    return;
  }
  setLoading(true);
  setError("");
  try {
    const body = JSON.stringify({
      origin: { lat: start.lat, lon: start.lon },
      destination: { lat: destination.lat, lon: destination.lon },
      alternatives: 3
    });
    let r = await fetch(`${API}/route/diverse`, {
      method: "POST",
      headers: { "Content-Type": "application/json" },
      body
    });
    if (!r.ok) r = await fetch(`${API}/route`, {
      method: "POST",
      headers: { "Content-Type": "application/json" },
      body
    });
    const d = await r.json();
    if (!r.ok) throw new Error(d.detail || "Route calculation failed.");
    setRoutes(d.routes || []);
    setSelectedRoute(0);
  } catch (e) {
    setError(e instanceof Error ? e.message : "Route calculation failed.");
  } finally {
    setLoading(false);
  }
```

```
};
const analyze = async (route: RouteOption) => {
  if (!vehicle) return;
  setLoading(true);
  setError("");
  try {
    const r = await fetch(`${API}/ev/analyze`, {
      method: "POST",
      headers: { "Content-Type": "application/json" },
      body: JSON.stringify({
        vehicle: vehicle.name,
        initial_soc_pct: soc,
        avg_speed_kmh: speed,
        ambient_temp_c: temp,
        weather, road_type: "Urban", soh_pct: soh,
        driving_style: style, route
      })
    });
    const d = await r.json();
    if (!r.ok) throw new Error(d.detail || "EV analysis failed.");
    setPlan(d);
  } catch (e) {
    setError(e instanceof Error ? e.message : "EV analysis failed.");
  } finally {
    setLoading(false);
  }
};
```

8.2 BACKEND API

Source file: api/main.py

FastAPI registers the route-generation and EV-intelligence modules used by the frontend.

Listing 8.2: API Initialization

```
app = FastAPI(
    title="Chennai EV Trip Planner API",
    version="0.1.0"
)
app.add_middleware(
    CORSMiddleware,
    allow_origins=[
        "http://localhost:5173",
        "http://127.0.0.1:5173"
    ],
    allow_credentials=True,
    allow_methods=["*"],
    allow_headers=["*"],
)
router = ChennaiRouter(GRAPH_PATH, BOUNDARY_PATH)
from api.ev_intelligence import router as ev_intelligence_router
```

```
app.include_router(ev_intelligence_router)
from api.route_diverse import register_diverse_routes
register_diverse_routes(app)
```

8.3 ROUTE PLANNING

Source file: api/route_diverse.py

Dijkstra's algorithm calculates the primary route using the configured travel-time edge weights. Edge penalties encourage alternative routes to use different road segments.

Listing 8.3: Dijkstra Route Calculation

```
def _dijkstra(compact, source, target, penalties):
    def weight(u, v, data):
        return data["base_time"] * penalties.get((u, v), 1.0)
    return nx.dijkstra_path(
        compact, source, target, weight=weight
    )
```

Listing 8.4: Alternative Route Generation — Core Logic

```
primary = _dijkstra(compact, source, target, {})
paths = [primary]
penalties = {}
for _ in range(1, count):
    for u, v in _path_edges(paths[-1]):
        penalties[(u, v)] = penalties.get((u, v), 1.0) * 6.0
    candidate = None
    for multiplier in (1.0, 2.0, 4.0, 8.0):
        trial = dict(penalties)
        if multiplier != 1.0:
            trial = {
                edge: value * multiplier
                for edge, value in trial.items()
            }
        try:
            trial_path = _dijkstra(
                compact, source, target, trial
            )
        except nx.NetworkXNoPath:
            continue
        if any(trial_path == p for p in paths):
            continue
        if max(
            (_path_overlap(trial_path, old) for old in paths),
            default=0.0
        ) <= 0.80:
            candidate = trial_path
            break
        if candidate is None:
            candidate = trial_path
```

```
    if candidate is None:
        break
    paths.append(candidate)
return paths
```

8.4 ENERGY PREDICTION

Source file: api/ev_intelligence.py

The prediction function prepares the EV and operating-condition features and uses the saved XGBoost model to estimate energy consumption in kWh.

Listing 8.5: Energy Prediction

```
def predict_energy(
    spec, distance_km, avg_speed, temp,
    weather, road_type, soh, style
):
    df = load_dataset()
    max_train_distance = float(
        pd.to_numeric(
            df["Distance_km"], errors="coerce"
        ).max()
    )
    ref_distance = 250.0
    base = {
        "Manufacturer": spec["Manufacturer"],
        "Model": spec["Model"],
        "Year": spec["Year"],
        "Battery_Capacity_kWh": spec["Battery_Capacity_kWh"],
        "Battery_Type": spec["Battery_Type"],
        "Motor_Power_kW": spec["Motor_Power_kW"],
        "Vehicle_Weight_kg": spec["Vehicle_Weight_kg"],
        "Avg_Speed": avg_speed,
        "Ambient_Temp_C": temp,
        "Weather": weather,
        "Road_Type": road_type,
        "SOH_pct": soh,
        "Driving_Style": style
    }
    model = load_model()
    def predict_at(distance):
        row = dict(base)
        row["Distance_km"] = float(distance)
        pred = float(model.predict(
            pd.DataFrame([row], columns=FEATURES)
        )[0])
        return max(0.0, pred)
    if distance_km <= max_train_distance:
        energy = predict_at(distance_km)
        meta = {"method": "direct_XGBoost"}
    else:
```

```
        ref_energy = predict_at(ref_distance)
        energy = (ref_energy / ref_distance) * distance_km
        meta = {
            "method": "XGBoost_reference_efficiency_scaling",
            "reference_distance_km": ref_distance
        }
    return energy, meta
```

8.5 SOC AND CHARGING OPTIMIZATION

Source file: api/ev_intelligence.py

The analysis function combines predicted energy with initial SOC and battery capacity. It estimates arrival SOC and safe range, then evaluates charging if the 25% reserve cannot be maintained.

Listing 8.6: SOC Estimation — Core Calculation

```
capacity = float(spec["Battery_Capacity_kWh"])
reserve_energy = capacity * RESERVE_SOC / 100.0
initial_energy = capacity * req.initial_soc_pct / 100.0
kwh_per_km = energy / max(route_distance, 1e-9)
arrival_soc = max(
    0.0,
    (initial_energy - energy) / capacity * 100.0
)
range_km = max(
    0.0,
    (initial_energy - reserve_energy) /
    max(kwh_per_km, 1e-9)
)
opt = optimize_single_stop(
    spec, req.initial_soc_pct, energy,
    route_distance, candidates
)
```

The charging optimizer uses binary station-selection variables and continuous charging quantities. It considers charging cost, charging time and route-access distance.

Listing 8.7: PuLP/MILP Objective and Constraints

```
problem = LpProblem(
    "EV_Charging_Stop_Optimization", LpMinimize
)
x = {
    f["idx"]: LpVariable(
        f"x_{f['idx']}", cat=LpBinary
    ) for f in feasible
}
q = {
    f["idx"]: LpVariable(
        f"q_{f['idx']}", lowBound=0
    ) for f in feasible
}
terms = []
```

```
for f in feasible:
    i = f["idx"]
    row = candidates.loc[i]
    price = float(row["price_rs_kwh"])
    power = max(1.0, float(row["power_kw"]))
    detour = float(row["distance_to_route_km"])
    terms.append(
        price * q[i]
        + 0.35 * (60.0 / power) * q[i]
        + 5.0 * detour * x[i]
    )
    problem += q[i] >= deficit * x[i]
    problem += q[i] <= f["max_charge"] * x[i]
problem += lpSum(x.values()) == 1
problem += lpSum(q.values()) >= deficit
problem += lpSum(terms)
problem.solve(PULP_CBC_CMD(msg=False))
```

8.6 APPLICATION OUTPUT

The implemented application displays the selected route, predicted energy consumption, estimated arrival SOC, safe range and charging status. When charging is needed and a feasible solution exists, it displays the recommended charging station and estimated charging details.

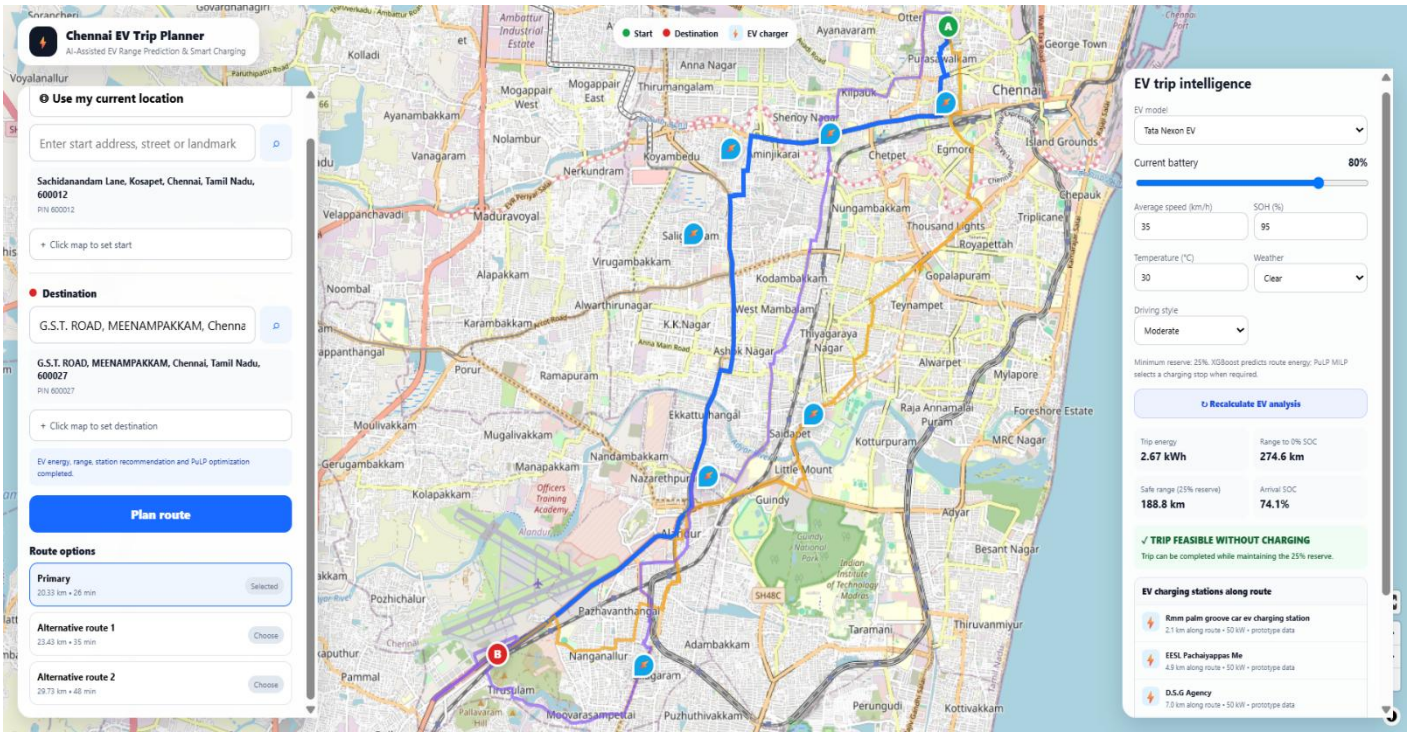


Figure 8.1. Implemented web application showing route selection and a trip-feasible result.

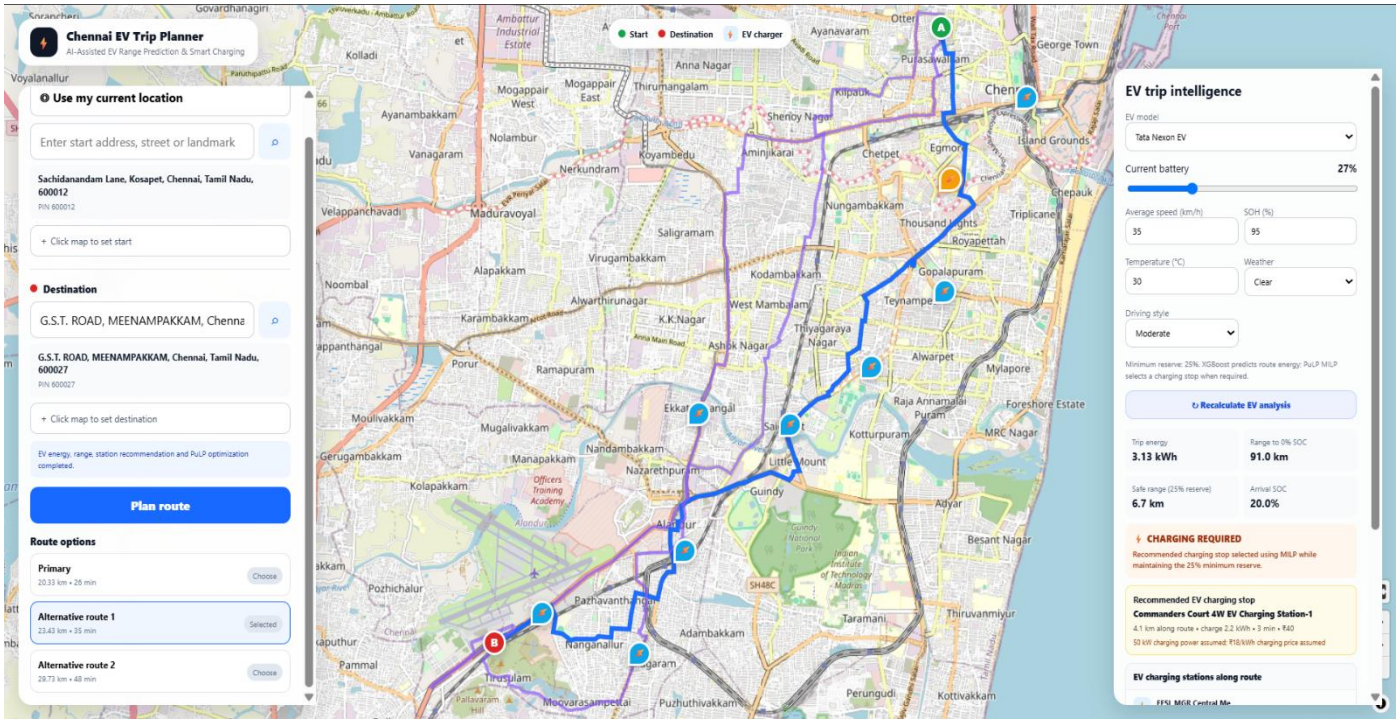


Figure 8.2. Low-SOC interface showing charging required and a recommended station.

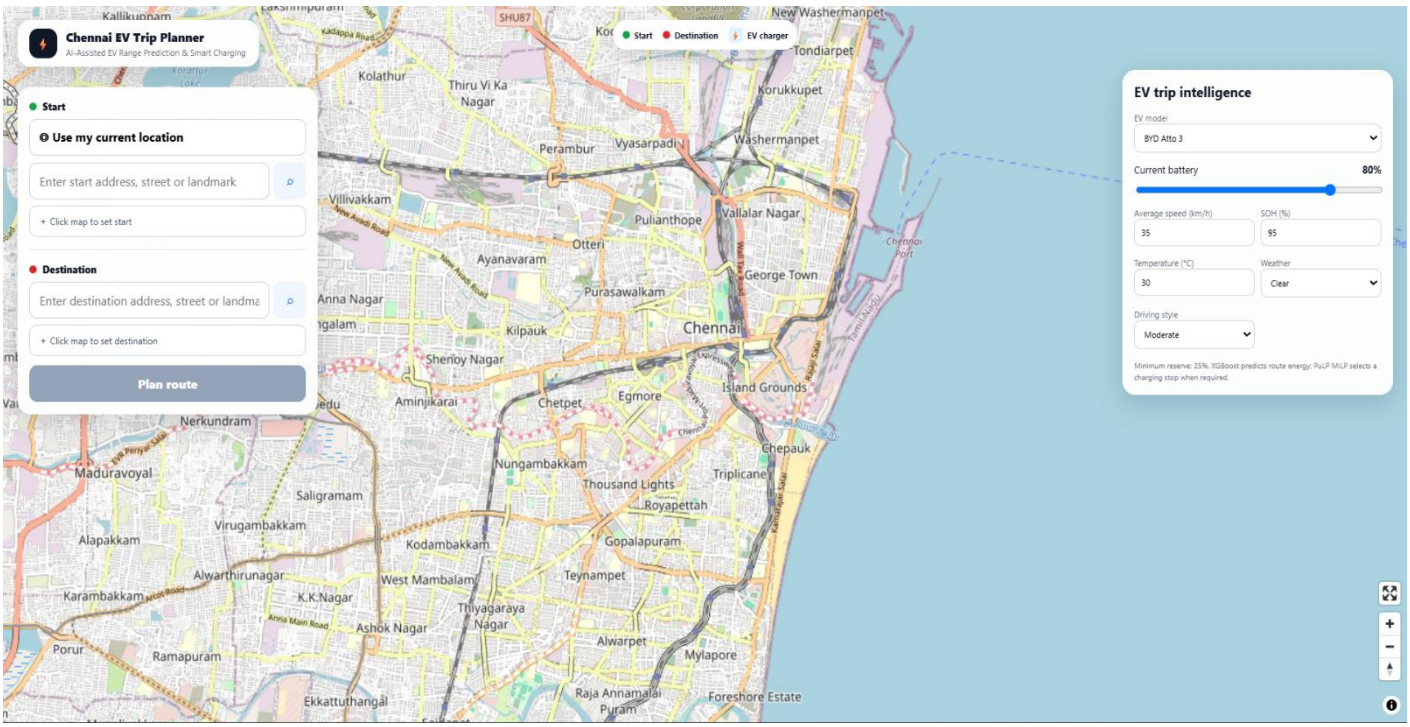


Figure 8.3. Initial trip-input interface before route planning.

8.7 SUMMARY

The application connects React and TypeScript with FastAPI, route generation, XGBoost energy prediction, SOC estimation and PuLP-based charging optimization. The final dashboard presents the results of the complete trip-planning workflow.

CHAPTER 9

CONFERENCE APPLICATION DETAILS

The work reported in this project was prepared as a conference paper and submitted through Microsoft Conference Management Toolkit (CMT). The submission details below are reproduced from the submission summary generated by the conference management system.

Table 9.1. Conference submission details

Field	Details
Submission system	Microsoft Conference Management Toolkit (CMT)
Conference name	International Conference on Computation, Communication, and Intelligent Systems in Engineering
Track name	Track 04: Smart Engineering Systems & Sustainable Infrastructure
Paper ID	664
Paper title	AI-Assisted EV Range Prediction and Smart Charging Optimization System
Submission status	Submission summary generated by the conference management system
Created on	07 October 2026, 18:32:48 (GMT+5:30)
Last modified	07 October 2026, 18:32:48 (GMT+5:30)
Primary subject area	Electric Vehicles & Energy Storage Systems
Submission file	IEEE-paper2.pdf
Authors	Sanjai S, Shoulina A, Madhumitha M and Anandan R, Vels Institute of Science, Technology & Advanced Studies (VISTAS), Chennai, India

Abstract Submitted to the Conference

Electric vehicle trip planning faces challenges due to uncertainty in energy use, battery levels, road conditions and charging needs. This paper introduces an AI-driven system for predicting EV range and optimizing charging. The system combines route planning, machine learning for energy prediction, battery state tracking and charging station selection using mathematical optimization. It works within Chennai District using browser-based location input, including geolocation, address search and map selection. A road network built from OpenStreetMap data enables route calculations. Dijkstra's algorithm finds the primary path, with alternative routes available for users to choose. An XGBoost regression model predicts energy use based on vehicle details, battery condition, road type, speed, weather and driving behaviour. The model output is used with the initial battery State of Charge (SOC) to estimate energy consumption, driving range and arrival SOC while maintaining a minimum 25% battery reserve. If a trip cannot maintain the required reserve, the system identifies reachable charging stations and uses a PuLP-based Mixed-Integer Linear Programming (MILP) model to determine charging decisions.

The system is implemented using React, MapLibre, FastAPI, MongoDB, Python, OSMnx, NetworkX, XGBoost and PuLP. Evaluation using 20,000 structured EV trip records produced an XGBoost test R^2 of 0.9978, MAE of 0.1481 kWh and RMSE of 0.3613 kWh.

Submission Summary

The conference submission concerns the integration of EV route planning, machine-learning-based energy prediction, battery SOC and range estimation, and reserve-constrained charging optimization. The paper describes the system implementation and its evaluation using the available dataset. The reported metrics represent model performance on the evaluation dataset and do not establish independent real-world fleet validation.

A copy of the conference submission summary is included as supporting evidence in this report.

CHAPTER 10

CONFERENCE APPLICATION / SUBMISSION EVIDENCE

The screenshot below is the submission-creation email received from Microsoft CMT for the paper. It is reproduced as submission evidence; it is not a notification of acceptance or publication.



Microsoft CMT <noreply@msr-cmt.org> Oct 7, 2026, 6:32 PM (13 days ago)

to me

Hello,

The following submission has been created.

Track Name: Track 04: Smart Engineering Systems & Sustainable Infrastructure

Paper ID: 664

Paper Title: AI-Assisted EV Range Prediction and Smart Charging Optimization System

Abstract:
Electric vehicle trip planning faces challenges to uncertainty in energy use, battery levels, road conditions and charging needs. This paper introduces an AI-driven system for predicting EV range and optimizing charging. The system combines route planning, machine learning for energy prediction, battery state tracking and charging station selection using mathematical optimization. It works within the Chennai District using browser-based tools for location input, like geolocation address search and map selection. A road network built from OpenStreetMap data enables route calculations. Dijkstra's algorithm finds the path with alternate routes available for users to choose. An XGBoost regression model predicts energy use based on vehicle details, battery condition, road type, speed, weather and driving behavior. The model's output is used with the starting battery state of charge to calculate energy, driving range and arrival SOC ensuring a minimum 25% battery reserve. If the trip exceeds the range the system identifies nearby charging stations that are reachable. It then uses a PuLP-based mixed-integer linear programming model to decide when and where to charge. The system is built with React, MapLibre, FastAPI, MongoDB, Python, OSMnx, NetworkX, XGBoost and PuLP. Testing with 20,000 real EV trip records showed performance: XGBoost achieved an R^2 of 0.9978, a MAE of 0.1481 and RMSE of 0.3613

Created on: Wed, 07 Oct 2026 13:02:48 GMT